

Vision Beyond Sight: AI-Enabled Smart Glasses for Visual Assistance Using Federated Learning

Sana Rahmani¹ and Rama Alkusair²

¹Department of Computer Science, Effat University, Jeddah, Saudi Arabia.

²Department of Computer Science, Effat University, Jeddah, Saudi Arabia.

Contributing authors: saurahmani@effat.edu.sa; ralkusair@effat.edu.sa;

Abstract

Visual impairment significantly limits independent navigation and scene understanding in daily life. Recent advances in vision-language models and wearable computing make it possible to deliver real-time, on-device visual assistance through smart glasses. This project presents a proof-of-concept system that combines image captioning and text-to-speech (TTS) with a federated-learning perspective for privacy-preserving model training. On the NLP side, we implement and fine-tune a pretrained VisionEncoderDecoder model (ViT-GPT2) on the Flickr8k image-caption dataset to generate natural-language descriptions of scenes captured by the smart-glasses camera. The fine-tuned captioning model is integrated with a TTS component to provide spoken feedback, enabling a complete image-to-speech pipeline suitable for visual assistance.

On the Advanced AI side, we address the privacy and scalability limitations of purely centralised training by studying how Federated Learning (FL) could be integrated into this setting. We propose a conceptual FL framework in which smart glasses act as clients that keep their images locally and only share model updates with a central server. Due to computational constraints, we implement a lightweight FL prototype in PyTorch using a small neural network and partitioned synthetic data to simulate multiple clients and FedAvg-based aggregation. Experimental results on this prototype illustrate how global performance can improve over successive communication rounds, while preserving data locality at the client side. Overall, the project delivers a working captioning-and-speech pipeline for smart glasses and lays the groundwork for future deployment of fully federated, privacy-aware visual assistance systems.

Keywords: smart glasses, visual assistance, image captioning, vision-language models, ViT-GPT2, text-to-speech, federated learning, FedAvg, privacy-preserving AI, assistive technology

1 Introduction

1.1 Background and Motivation

Visual impairment and low vision significantly reduce a person’s ability to perceive and understand their surroundings, making everyday tasks such as navigation, object recognition, and reading textual information highly challenging. Recent advances in wearable computing and artificial intelligence (AI) have enabled the development of smart glasses that integrate cameras, on-device processing, and audio output to provide real-time visual assistance. Commercial products such as Envision Glasses and other AI-enabled smart glasses can already read text aloud, recognize objects and faces, and describe the environment to blind and low-vision users, illustrating the practical value of combining computer vision and speech technologies in a wearable form factor [1, 2].

At the same time, progress in vision-language modelling has led to powerful architectures that jointly process images and natural language. Image captioning models trained on datasets such as Flickr8k, which contains around 8,000 images paired with multiple human-written captions, are widely used as benchmarks for learning to generate descriptive sentences from visual input [3]. Transformer-based encoder-decoder models, such as the ViT-GPT2 VisionEncoderDecoder architecture, combine a Vision Transformer (ViT) encoder with a GPT-2 language decoder to produce fluent descriptions of images and have been shown to perform strongly on standard captioning datasets [4]. When coupled with text-to-speech (TTS) components, such models form a complete image-to-speech pipeline that can be embedded into smart glasses to support scene understanding for visually impaired users.

However, many existing assistive systems rely on centralised cloud processing, where images captured by the glasses are uploaded to remote servers for analysis. This raises serious privacy concerns, as visual data from smart glasses may contain highly sensitive information about the user’s home, social interactions, documents, and physical appearance. In addition, purely centralised training of large vision-language models typically assumes that all data are collected and stored in a single data centre, which does not align well with scenarios in which data are naturally distributed across many personal devices.

1.2 Problem Statement

The central problem addressed in this project is how to design and study an AI-enabled smart-glasses system for visual assistance that balances three requirements:

- generating meaningful, natural-language descriptions of a user’s visual surroundings that can be spoken aloud in real time;

- operating under realistic computational constraints typical of edge or wearable devices;
- and respecting the privacy of users’ visual data by reducing the need to centralise raw images during model training and improvement.

From a modelling perspective, this translates into two intertwined challenges. First, we need an effective image captioning component that can be fine-tuned on a benchmark dataset and integrated with TTS to form a working prototype of a smart-glasses assistance pipeline. Second, we need to explore how Federated Learning (FL)—a paradigm in which multiple clients collaboratively train a shared model without exchanging raw data—could be conceptually integrated into such a system to allow future deployments to improve models across many devices while keeping visual data local [5–7]. Addressing these challenges requires combining techniques from natural language processing (NLP), computer vision, and advanced AI training strategies.

1.3 Research Questions

Based on the above problem statement, this project is guided by the following research questions:

- **RQ1:** How effectively can a fine-tuned vision–language image captioning model describe real-world scenes in a way that is suitable for delivering spoken visual assistance through smart glasses?
- **RQ2:** How can a conceptual Federated Learning framework be designed for AI-enabled smart glasses so that multiple devices can collaboratively improve a shared model without uploading raw visual data to a central server?
- **RQ3:** What opportunities and challenges are highlighted in recent literature regarding the use of Federated Learning and edge AI for privacy-preserving, vision-based assistive or wearable systems?

1.4 Project Objectives

In order to address the problem outlined above and to support both the NLP and Advanced AI components of the project, we define the following high-level objectives:

- **O1:** Investigate and synthesise recent research on AI-enabled assistive technologies and Federated Learning for vision and wearable systems, with a particular focus on privacy-preserving, edge-centric designs for visually impaired users.[6–8]
- **O2:** Develop a conceptual framework for integrating Federated Learning into an AI-enabled smart-glasses ecosystem, specifying the roles of clients and server, the flow of model updates, and the main privacy and deployment considerations for collaborative training.
- **O3:** Implement a lightweight Federated Learning prototype using a small neural network and partitioned data, in order to gain practical insight into the behaviour of federated optimisation algorithms under realistic computational and communication constraints.

- **O4:** Design and implement a proof-of-concept vision–language pipeline that converts images captured in a smart-glasses setting into spoken natural-language descriptions, by fine-tuning an appropriate image captioning model on representative caption data and coupling it with a text-to-speech component.
- **O5:** Provide a preliminary system-level and cost-oriented analysis of a potential smart-glasses deployment, outlining the hardware, software and infrastructure stack required to support both the captioning pipeline and a future federated training setup.

Taken together, these objectives ensure that the project not only delivers a working image-to-speech assistance pipeline, but also situates it within a broader investigation of federated, privacy-aware training strategies and the practical constraints of implementing such systems on real-world smart-glasses platforms.

1.5 Scope and Limitations

The scope of this project is intentionally constrained to make the study feasible under the time and computational resources available. On the NLP side, we focus on a single vision–language architecture, namely the pretrained ViT–GPT2 VisionEncoderDecoder model, which is fine-tuned on the Flickr8k dataset and evaluated on held-out images [3, 4]. The implementation targets a proof-of-concept software pipeline rather than a complete hardware prototype, and we do not perform real-time optimisation or deployment on an actual smart-glasses device.

On the Federated Learning side, we do not train the full ViT–GPT2 captioning model in a realistic federated setting, as this would exceed the computational capacity of the available environment. Instead, we adopt a two-layer strategy. At the conceptual level, we design and discuss how FL could be integrated into a smart-glasses ecosystem, in line with established FL formulations and recent surveys on privacy-preserving edge AI [5–7, 9]. At the experimental level, we implement a simplified FL prototype in PyTorch using a small neural network and synthetic or partitioned data to emulate multiple clients and server-side FedAvg aggregation. This prototype is used solely to study the behaviour of FL algorithms and is not claimed to be a full deployment of federated training for the captioning model. The overall contribution of the work should therefore be viewed as a combination of (i) a practical captioning-and-speech pipeline suitable for visual assistance, and (ii) foundational steps towards a federated, privacy-aware training strategy for future smart-glasses systems.

2 Literature Review

2.1 Assistive Technologies and AI-Enabled Smart Glasses for Visual Assistance

Visual impairment affects hundreds of millions of people worldwide and creates persistent barriers in education, mobility, social interaction, and employment. Traditional assistive technologies—such as the white cane, Braille books, magnifiers and basic electronic aids—have been essential for decades but remain limited in terms of adaptability, contextual understanding and personalization.[8, 10] Recent reviews show a

clear shift from purely mechanical or low-level electronic tools toward AI-powered systems that can interpret rich visual scenes, read text in real time and support multimodal interaction.[8, 11]

From a functional perspective, modern assistive solutions for visually impaired users can be grouped into three major categories: (i) computer-vision based systems that perform object detection, scene description and text recognition; (ii) natural language processing (NLP) and text-to-speech (TTS) technologies that convert digital or captured text into speech and enable voice-driven interaction; and (iii) wearable devices that integrate sensors, edge AI and audio feedback into a portable form factor.[8, 10] Popular smartphone applications such as Microsoft Seeing AI and Google Lookout demonstrate the potential of cloud-based computer vision for reading documents, identifying products and providing scene descriptions, but they depend heavily on network connectivity and remote servers for inference.[10]

Within this landscape, AI-enabled smart glasses have emerged as one of the most promising wearable platforms for visual assistance. These systems typically combine a head-mounted camera, on-board or companion-compute (e.g., smartphone or edge device), and an audio interface to deliver real-time descriptions of the user’s environment.[8, 12] Commercial devices such as OrCam MyEye and Envision Glasses provide functionalities including text reading, object recognition, barcode scanning and face recognition, all delivered as spoken feedback to the user.[8, 13] Recent engineering prototypes extend these capabilities with obstacle detection, currency recognition and multilingual OCR, often using deep learning models (e.g., YOLO for object detection and CNN-based or transformer-based networks for scene understanding) running on embedded platforms like Raspberry Pi or ESP32.[14–16]

Clinical and user studies confirm that smart glasses can substantially enhance independence when the hardware and user interface are well designed. For example, Udayakumar et al. evaluated a spectacle-mounted “Smart Vision Glass” (SVG) in a multicenter Indian cohort of 90 participants with severe visual impairment.[17] The device offered four core functions—“Things Around You”, “Reading”, “Walking Assistance” and “Face Recognition”—and all participants were able to access every function after training. The most frequently used and highly rated features were reading and environmental awareness, indicating that real-time text access and object description directly address everyday challenges such as studying, navigating indoor environments and identifying people.[17] Similar prototypes reported in the engineering literature combine scene description, OCR and obstacle detection and achieve object and text recognition accuracies in the range of 75–95% under controlled conditions, highlighting the technical feasibility of smart-glass based assistance.[14, 15]

At the same time, recent systematic and bibliometric reviews point out that current wearable mobility aids and smart glasses still face important limitations.[11, 12, 18] First, many commercial systems are expensive and thus inaccessible for users in low- and middle-income settings.[8, 17] Second, a large proportion of AI functionality relies on cloud-based processing, which raises privacy concerns (because images of the user’s surroundings are transmitted to remote servers) and makes the device unusable when wireless connectivity is poor or unavailable.[8, 10] Third, language and cultural coverage remain limited: support for non-dominant languages, local currencies and

region-specific objects is often incomplete.[8, 15] Finally, several reviews emphasise that many proposed systems are evaluated only in small-scale or laboratory settings, with limited long-term usability studies and minimal involvement of visually impaired users in the design process.[11, 12]

These gaps have motivated a growing interest in privacy-preserving and edge-centric assistive systems. Mekonnen et al. demonstrate that a fully offline platform based on Raspberry Pi 5 can integrate object detection (YOLOv8), OCR, face recognition and voice-command control while keeping all data on-device, thus avoiding cloud transmission and reducing latency.[10] In parallel, several design-oriented papers on smart glasses argue for user-centred, low-cost architectures that combine embedded AI with efficient audio feedback and secure connectivity only when necessary.[14, 16]

In our project, we position the proposed AI-enabled smart glasses within this broader evolution of assistive technologies. We focus specifically on a vision-language image captioning pipeline that converts camera images into natural-language descriptions and then into speech, aiming to provide a flexible layer of semantic understanding that complements traditional OCR or fixed object categories. At the same time, we adopt a long-term perspective in which such smart glasses will be coupled with federated learning to update models collaboratively across users without centralising raw image data, thereby addressing both the privacy and adaptability limitations identified in recent literature.[8, 10, 18]

2.2 Image Captioning for Visual Assistance

Image captioning is the task of automatically generating a natural-language description for a given image. It sits at the intersection of computer vision and natural language processing, requiring the model to detect salient objects, reason about their attributes and relationships, and then produce a grammatically correct and semantically coherent sentence.[19, 20] Early work framed image description as a ranking problem: given an image and a set of candidate captions, the model learns a joint visual-textual embedding space where correct image-caption pairs are closer than mismatched pairs. The Flickr8k dataset introduced by Hodosh et al.—with 8,000 images and five human-written captions per image—became a widely used benchmark for this line of research and remains a standard dataset for evaluating modern captioning models.[3]

With the rise of deep learning, encoder-decoder architectures became dominant in image captioning.[19] In the influential “Show and Tell” model, a convolutional neural network (CNN) such as Inception or ResNet is used as an image encoder to produce a fixed-length feature vector that is fed into a recurrent neural network (RNN)—typically an LSTM—which generates a caption word by word.[21] This approach allowed end-to-end training from raw pixels to sentences and achieved strong performance on Flickr8k, Flickr30k and MS COCO. Subsequent work introduced attention mechanisms to let the decoder focus selectively on different regions of the image at each time step. Xu et al.’s “Show, Attend and Tell” model learns a soft visual attention map so that the captioning network can “look” at different parts of the image when generating different words, improving both accuracy and interpretability.[22] Later, Anderson et al. proposed bottom-up and top-down attention that operates at

the level of object proposals (e.g., from Faster R-CNN) and achieved state-of-the-art results on MS COCO captioning metrics.[23]

Comprehensive surveys of image captioning techniques show a clear evolution from template-based and retrieval-based approaches toward deep encoder-decoder architectures with increasingly sophisticated attention and fusion mechanisms.[19, 20] These surveys also emphasise the role of benchmark datasets and evaluation metrics. Flickr8k, Flickr30k and MS COCO are the most commonly used datasets, each providing multiple human captions per image, which allows models to be assessed against diverse human descriptions.[3, 20] Traditional automatic evaluation relies on n-gram overlap metrics such as BLEU, METEOR, ROUGE-L and CIDEr, and more recently on SPICE and learned metrics that correlate better with human judgements.[19, 24] Recent work further argues that as captions become longer and more diverse, rule-based metrics may struggle to capture semantic adequacy, motivating the use of learned or multimodal evaluators.[24]

In parallel with CNN-RNN architectures, transformer-based models have reshaped the image captioning landscape. Vision Transformers (ViT) treat images as sequences of patches processed by self-attention, while autoregressive language models such as GPT-2 generate text token by token.[19] The VisionEncoderDecoder framework used in the Hugging Face ecosystem generalises this idea by allowing a pre-trained vision encoder (e.g., ViT) to be combined with a pre-trained text decoder (e.g., GPT-2) into a single image-to-text model.[25] The `nlpconnect/vit-gpt2-image-captioning` model is a concrete instance of this family: it couples a ViT encoder with a GPT-2 decoder and is trained for image captioning on datasets such as MS COCO.[4] Public documentation and tutorials describe how this architecture can be fine-tuned on smaller datasets like Flickr8k by preparing an image-caption CSV file, tokenising captions, and optimising the model using standard sequence-to-sequence training objectives.[4, 25] This is precisely the strategy adopted in our project, where we fine-tune the ViT-GPT2 VisionEncoderDecoder model on Flickr8k to tailor captioning behaviour to our smart-glasses scenario.

Beyond general benchmarks, several recent efforts explicitly target image captioning as an assistive technology. The VizWiz-Captions dataset collects more than 39,000 images taken by blind photographers with mobile phones, each paired with five human captions, in order to model the real visual conditions experienced by blind users (e.g., blur, occlusion, poor framing).[26] Yu et al. argue that images captured by visually impaired people are often noisy and that standard captioning models can produce unsafe or misleading descriptions in these conditions; they propose a quality-agnostic framework that uses noise-aware data augmentation, dual-network architectures and consistency losses to improve robustness, achieving measurable gains on captioning metrics for low-quality images.[27] Other work focuses on designing captioning systems for low-resource languages and embedded deployment: for instance, a recent TechRxiv preprint describes an image-captioning-and-TTS pipeline tuned for visually impaired users in low-resource language settings, optimised with TensorRT for faster inference on constrained hardware.[28]

Industrial and applied research further reinforces the link between image captioning and accessibility. IBM’s “Science for Social Good” initiative has highlighted

image captioning as a key assistive technology, stressing that automatically describing user-generated images can help blind people better understand photos they take or receive.[29] The VizWiz challenge and related competitions have similarly encouraged the community to develop captioning models that are robust to real-world image artefacts and that prioritise safety and usefulness over purely aesthetic descriptions.[26, 27] Together, these works suggest that high-quality captioning for visual assistance must address three intertwined requirements: accurate recognition of salient visual content, robustness to noisy and atypical images, and generation of concise, user-centred descriptions.

Within this body of literature, our project is positioned as a practical instantiation of transformer-based image captioning for assistive use. By fine-tuning a ViT-GPT2 VisionEncoderDecoder model on Flickr8k and integrating it with a text-to-speech (TTS) module, we implement an image-to-speech pipeline that can describe scenes captured by smart glasses in natural language. Rather than proposing a new captioning architecture, we leverage a well-established model and training recipe and focus on: (i) adapting it to our dataset and scenario; (ii) evaluating its behaviour qualitatively on unseen images; and (iii) treating it as the NLP backbone for a future federated-learning setup, where model updates would be derived from user-specific visual data without centralising raw images.

2.3 Federated Learning for Privacy-Preserving Vision and Wearable AI

2.3.1 Core Concepts of Federated Learning

Federated Learning (FL) can be understood as a generalisation of distributed optimisation in which the objective is to minimise a global loss function over data that are partitioned across many clients, without ever centralising those data.[5, 30] Formally, let there be K clients, where client k holds a local dataset $\mathcal{D}_k$ of size n_k . The overall learning objective is to find model parameters w that minimise the weighted sum of local empirical losses:

$$\min_w F(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w), \quad (1)$$

where $n = \sum_{k=1}^K n_k$ is the total number of samples and $F_k(w)$ is the empirical loss on client k . In classical centralised training, all data $\{\mathcal{D}_k\}$ would be uploaded to a single server to optimise $F(w)$ directly. In FL, by contrast, each client keeps $\mathcal{D}_k$ locally and only model parameters or updates are communicated.[5, 30]

The most widely used algorithm in this setting is Federated Averaging (FedAvg).[5] Training proceeds in rounds $t = 0, 1, 2, \dots$ under the control of a central server:

1. **Broadcast:** The server maintains a global model $w^{(t)}$ and sends it to a subset S_t of participating clients selected for round t (e.g., uniformly at random).
2. **Local training:** Each client $k \in S_t$ initialises its local model with $w^{(t)}$ and performs one or more epochs of stochastic gradient descent (SGD) on its own dataset $\mathcal{D}_k$, obtaining an updated local model $w_k^{(t+1)}$.

3. **Aggregation:** Each client sends its updated parameters (or the difference $w_k^{(t+1)} - w^{(t)}$) back to the server. The server aggregates these updates by computing a weighted average:

$$w^{(t+1)} = \sum_{k \in S_t} \frac{n_k}{\sum_{j \in S_t} n_j} w_k^{(t+1)}. \quad (2)$$

This iterative process continues until convergence or until a fixed number of rounds is reached. FedAvg can be seen as a form of local-SGD in which each client carries out multiple local optimisation steps before synchronising with the server, thus reducing the frequency of communication compared to traditional synchronous distributed SGD.[5, 30]

Kairouz et al. characterise FL systems along three main axes: *cross-device* vs. *cross-silo*, *horizontal* vs. *vertical* data partitioning, and *federated* vs. more general decentralised learning.[30] In cross-device FL, the number of clients is very large (e.g., millions of smartphones or wearables), individual datasets are small, and client availability is highly variable. In cross-silo FL, the number of clients is modest (e.g., tens of hospitals or companies), each with a relatively large and stable dataset. In horizontal FL, clients share the same feature space but have different samples (e.g., different patients or images), whereas in vertical FL, different institutions hold different feature subsets for the same set of entities (e.g., a bank and a hospital each holding complementary attributes about overlapping individuals).[30] The smart-glasses scenario studied in this project naturally fits the cross-device, horizontal FL setting: many end-user devices, each with its own stream of images, all learning a shared captioning model.

From a privacy and security perspective, Truong et al. emphasise that FL should not be viewed as a complete privacy solution on its own, but rather as a building block within a broader *privacy-by-design* architecture.[31] Even if raw data never leave the device, model updates can leak information through gradient-based attacks or membership inference, especially when models are over-parameterised or client datasets are small.[30, 31] Consequently, many FL systems integrate additional mechanisms such as:

- **Secure aggregation**, where cryptographic protocols ensure that the server only observes the sum of client updates, not each update individually.
- **Differential privacy**, where calibrated noise is added to updates or to the aggregated model to provide formal privacy guarantees with respect to individual training examples.
- **Robust aggregation**, where the server employs aggregation rules designed to mitigate the influence of malicious or corrupted clients (e.g., median-based or trimmed-mean aggregation).

Surveys focused on FL for IoT and healthcare stress that choosing an appropriate combination of these techniques is crucial when dealing with sensitive streams such as medical images, physiological signals or continuous audio-visual recordings.[32, 33]

Finally, modern FL research increasingly views FL not only as a training mechanism but also as a system-design paradigm.[30, 34] Decisions such as how often to trigger FL rounds, how many local epochs to run, which clients to sample in each

round, and how to offload parts of the computation to edge servers all fall under the umbrella of FL system design. For edge and wearable devices, these trade-offs must be balanced against constraints on battery life, computation, memory and connectivity. In the context of AI-enabled smart glasses for visual assistance, this means that any future FL deployment would need to carefully schedule training rounds (e.g., when the device is charging and connected to Wi-Fi), select appropriate model sizes and consider whether part of the computation should be offloaded to nearby edge nodes. The conceptual framework developed in this project is informed by these core FL concepts and is tailored to the cross-device, privacy-sensitive nature of smart-glasses data.

2.3.2 Federated Learning in Vision, Medical Imaging and Smart Healthcare

Due to the sensitivity and distributed nature of medical and vision data, FL has attracted particular attention in medical imaging and smart healthcare. Guan et al. present a comprehensive survey of FL for medical image analysis, covering classification, segmentation and detection tasks in modalities such as MRI, CT and histopathology.[35] They show that FL can often achieve performance close to centralised training when appropriate aggregation strategies, learning-rate schedules and client sampling schemes are used, while enabling hospitals and clinics to avoid sharing raw patient scans. The survey also highlights technical open problems highly relevant to vision-based systems: handling domain shifts between institutions, mitigating the impact of non-IID label distributions, and designing communication-efficient protocols for large 3D images.[35]

Nguyen et al. review FL for smart healthcare from a broader systems perspective, focusing on Internet-of-Medical-Things scenarios in which data originate not only from hospitals but also from wearables and home monitoring devices.[36] They classify FL methods into resource-aware, secure and privacy-aware, incentive-based and personalised variants, and discuss applications in remote health monitoring, electronic health records and medical imaging. Abbas et al. similarly analyse FL in smart healthcare, emphasising that FL is particularly attractive when sensor data streams are privacy-sensitive, high-volume and generated continuously at the network edge.[33] These works collectively suggest that FL is well suited for scenarios where visual or physiological signals from many individuals must be leveraged to improve a shared model without aggregating all data in a central repository.

Although most existing FL deployments in healthcare focus on supervised learning for classification or segmentation, the underlying principles extend naturally to vision-language tasks. For example, a federated medical imaging system might train a segmentation network across multiple hospitals without sharing raw scans, while a federated assistive-vision system could fine-tune an image captioning model across multiple users' devices without uploading their personal images.[35, 36] In both cases, the model parameters become a shared resource that encodes population-level patterns, while each client retains control over its raw data and can influence the global model through local updates.

2.3.3 Federated Learning for Edge, IoT and Wearable Devices

Beyond healthcare, FL has been widely explored in edge and IoT environments, where billions of devices generate data that are privacy-sensitive, bandwidth-limited or both.[32, 34] Zhan et al. discuss cloud–edge–end collaborative FL architectures in which end devices (e.g., phones, cameras, wearables) send model updates to nearby edge servers, which in turn coordinate with a cloud server for global aggregation.[34] Li et al. propose a cloud–edge–end collaborative FL method that explicitly tackles non-IID data by grouping edge nodes according to data similarity and incorporating generative models to mitigate distribution imbalance, achieving improved accuracy and F1-score compared to standard FedAvg on real-world datasets.[37] Other work on end–edge–cloud FL shows that distributing different parts of the training workload across tiers can alleviate the resource constraints of low-power devices while still providing strong privacy guarantees.[38]

Wearable devices are increasingly mentioned in these surveys as natural FL clients: smartwatches and fitness bands continuously record physiological signals; AR/VR headsets and smart glasses capture rich visual and spatial information.[32, 33] In many cases, centralising all such data would be impractical or unethical, especially when recordings include bystanders or intimate contexts. FL offers a path for training and updating models that run on wearables while keeping raw sensor streams local to each user. For camera-based wearables specifically, the combination of high-dimensional images, strong privacy expectations and potential connectivity limitations makes federated and edge-centric training strategies particularly relevant.

2.3.4 Positioning the Present Work

Within this broader FL landscape, the present project adopts a deliberately scoped approach. Rather than attempting to train a large vision–language captioning model in a full-scale federated setting—which would exceed the computational resources available in an academic environment—we separate our contribution into two complementary layers. At the *NLP layer*, we implement and fine-tune a ViT-GPT2 VisionEncoderDecoder model for image captioning on the Flickr8k dataset and integrate it with a text-to-speech component to create an image-to-speech pipeline suitable for smart-glasses visual assistance. At the *FL layer*, we study how a federated training paradigm could be integrated into this ecosystem by designing a conceptual framework in which smart glasses act as FL clients that retain their local image data while periodically participating in federated training rounds.

Practically, we instantiate this FL layer via a lightweight prototype: a small neural network trained in a simulated FL setting with partitioned data and FedAvg-based aggregation. This prototype is not intended as a final model for caption generation; instead, it serves as a concrete vehicle for exploring the behaviour of FL algorithms, including the impact of client sampling, local epochs and communication rounds on global performance.[5, 32] Conceptually, the framework aligns with recent FL surveys that call for privacy-preserving, edge-centric AI in IoMT and wearable scenarios, where devices such as smart glasses continually capture sensitive visual information.[31, 34, 36] By combining a working centralised captioning-and-speech

pipeline with a conceptual and prototypical treatment of FL, the project aims to lay methodological groundwork for future smart-glasses systems in which federated, privacy-aware training of vision–language models becomes feasible at scale.

2.4 Summary and Research Gaps

The literature reviewed in this section highlights three complementary strands of work relevant to our project. First, research on AI-powered assistive technologies and smart glasses shows that deep-learning based computer vision and speech technologies can substantially improve independence for blind and low-vision users.[8, 12, 17] Commercial and research prototypes already offer functions such as text reading, object and face recognition, and basic scene understanding. However, several limitations recur across these studies: many systems depend on cloud-based processing and continuous network connectivity, raising privacy and reliability concerns; device cost and proprietary hardware designs restrict accessibility; and long-term, real-world evaluations with visually impaired users remain relatively scarce.[8, 10, 18] Recent work on fully offline assistive platforms illustrates that on-device deep learning is technically feasible, but most existing designs are task-specific (e.g., OCR plus object detection) and do not fully exploit richer semantic descriptions of scenes.[10, 14]

Second, the image captioning literature has evolved from early CNN–RNN encoder–decoder models toward transformer-based vision–language architectures.[19, 20] Benchmark datasets such as Flickr8k, Flickr30k and MS COCO, together with metrics like BLEU, METEOR and CIDEr, have enabled systematic progress and rigorous evaluation.[3, 20] Recent work explores more robust captioning models tailored to images captured by blind photographers (e.g., VizWiz-Captions) and investigates quality-agnostic frameworks that remain useful under noisy, low-quality conditions.[26, 27] At the same time, the emergence of VisionEncoderDecoder models that combine a ViT encoder with a GPT-style decoder, such as the `vit-gpt2-image-captioning` model, has made it comparatively easy to fine-tune strong captioning baselines for downstream applications.[4, 25] Nonetheless, relatively few works integrate such transformer-based captioning pipelines into wearable smart-glasses prototypes, especially in a way that explicitly targets visually impaired users and connects captioning to broader system and privacy considerations.[8, 28]

Third, the federated learning (FL) literature provides a rich set of concepts and tools for training models across distributed, privacy-sensitive datasets without centralising raw data.[5, 30] Surveys in healthcare, IoMT and edge computing demonstrate that FL can approach the performance of centralised training for classification and segmentation tasks while keeping patient data local, but they also highlight challenges related to non-IID data, resource constraints and privacy leakage through model updates.[32–36] Wearable devices, including camera-equipped systems, are repeatedly identified as natural FL clients, yet most existing deployments focus on relatively low-dimensional signals (e.g., time-series physiological data) or on vision models that perform classification rather than full image-to-text generation.[32, 33]

Taken together, these strands reveal a set of research gaps that motivate the present work. There is a need for assistive smart-glasses prototypes that (i) leverage

modern transformer-based image captioning to provide richer, semantically meaningful scene descriptions; (ii) are designed with a clear path toward on-device or edge-centric deployment, rather than relying exclusively on cloud inference; and (iii) are conceptually aligned with federated learning, so that future versions can improve their vision–language models collaboratively across many users without uploading raw visual data.[8, 10, 30, 34] Our project addresses these gaps by combining a practical ViT–GPT2-based captioning-and-speech pipeline with a federated-learning perspective tailored to the smart-glasses scenario, and by situating this combination within a broader discussion of system architecture and cost in subsequent sections.

3 Conceptual Framework and System Architecture

3.1 Smart-Glasses AI Pipeline: Image-to-Speech Assistance

The core AI functionality of the proposed smart-glasses system is an image-to-speech pipeline that converts visual scenes captured by a head-mounted camera into spoken natural-language descriptions. Conceptually, this pipeline consists of four main stages: (i) image capture and basic pre-processing; (ii) vision–language caption generation; (iii) text-to-speech (TTS) synthesis; and (iv) audio delivery to the user through a wearable audio interface.

Image capture and pre-processing.

In the envisioned deployment, a miniature camera embedded in the glasses continuously or intermittently captures RGB frames from the user’s field of view. For the purposes of this project, we emulate this behaviour using still images stored in a dataset (Flickr8k) and sample images outside the training split to mimic unseen scenes. In a real-time prototype, frames would be sampled either at fixed intervals or in response to user input (e.g., a button press or voice command) to avoid overwhelming the user with constant output. Before being fed to the captioning model, each image is resized and normalised according to the requirements of the chosen Vision Transformer (ViT) encoder (e.g., resizing to a fixed resolution and scaling pixel values to the range expected by the pre-trained model).[4, 25]

Vision–language caption generation.

The central component of the pipeline is a vision–language model that produces a natural-language description from the processed image. In our implementation, we adopt a VisionEncoderDecoder architecture that couples a ViT image encoder with a GPT-2 language decoder, instantiated via the `nlconnect/vit-gpt2-image-captioning` model available in the Hugging Face ecosystem.[4] The ViT encoder maps the input image into a sequence of visual embeddings, which are then consumed by the GPT-2 decoder to autoregressively generate a caption token by token. During training, the model is fine-tuned on the Flickr8k dataset using a standard sequence-to-sequence loss that encourages the decoder to reproduce human-written captions. During inference on the smart-glasses pipeline, the model is run in generation mode, using beam search with a limited maximum length to produce concise descriptions that are appropriate for real-time audio output.[19, 20]

Text post-processing and interaction design.

The raw caption produced by the model may contain artifacts such as repeated characters, overly long sentences or uncertain phrases. In our prototype, simple post-processing rules are applied to clean the text, such as collapsing repeated characters, trimming redundant punctuation and enforcing an upper bound on caption length. In a full system, more sophisticated post-processing could incorporate user preferences (e.g., level of detail, language, or verbosity) and context (e.g., current task, environment type). Interaction design is also crucial: the system must decide when to describe a scene, how frequently to update the description as the user moves, and how to avoid interrupting the user with redundant or unhelpful messages. While these aspects are outside the primary scope of the current implementation, they form an important part of the conceptual architecture for a deployable smart-glasses assistant.[8, 17]

Text-to-speech (TTS) synthesis.

Once a caption string is produced and optionally post-processed, it is passed to a text-to-speech module that converts the text into an audio waveform. In our software prototype, we employ a standard neural or cloud-based TTS engine (e.g., a library such as gTTS or an equivalent on-device TTS service) to synthesise speech in English. In a realistic deployment, the TTS component would need to support multiple languages and voices, low latency and high intelligibility in noisy environments.[18] Modern TTS systems based on sequence-to-sequence or neural vocoder architectures (e.g., Tacotron-2, FastSpeech, WaveNet-style vocoders) can produce natural-sounding speech and are widely integrated into mobile and embedded platforms.[39, 40]

Audio delivery to the user.

The final stage of the pipeline is audio playback through a wearable interface such as bone-conduction transducers or open-ear speakers integrated into the glasses frame. These technologies allow the user to receive spoken feedback while preserving awareness of environmental sounds, which is critical for mobility and safety.[8, 17] The audio module must provide sufficient volume and clarity without causing discomfort during prolonged use. In addition, the system should support basic controls (e.g., repeat last description, pause, change speed) through buttons or voice commands. Although the present project does not implement a full hardware prototype, these considerations guide the conceptual design of the smart-glasses pipeline and inform later discussions of hardware options and cost in Section 3.2.

Overall, this image-to-speech pipeline serves as the NLP backbone of the proposed system: it defines how visual input is transformed into spoken output in a way that can support everyday tasks for visually impaired users. In subsequent sections, we build on this backbone by (i) specifying how Federated Learning could be integrated to update the captioning model collaboratively across many devices without sharing raw images, and (ii) outlining a realistic hardware and software stack, together with indicative costs, for a future smart-glasses prototype based on this architecture.

Link to the implemented prototype and scope limitations.

In the present project, practical constraints prevent us from running full-scale FL on the ViT-GPT2 captioning model itself or deploying FL on real smart-glasses hardware. Instead, the FL contribution is deliberately scoped to two complementary components. First, we develop a conceptual client-server architecture in which AI-enabled smart glasses act as FL clients that retain their local image data and periodically contribute model updates, while a coordinating server performs privacy-aware aggregation. This framework is grounded in established FL literature on cross-device, horizontal settings and edge-centric deployments.[5, 30, 34] Second, we implement a lightweight FL simulation using a small neural network and partitioned data on a single machine, in order to study core FL behaviours such as the impact of client sampling, local epochs and communication rounds on global performance under realistic resource constraints.[5, 32]

This design choice is intentional: our aim is not to claim a fully deployed federated captioning system, but rather to provide a rigorous conceptual and experimental foundation that could be extended in future work if additional computational resources and hardware support become available. In the context of a senior design project, this means that the same FL framework and simulation strategy can be further refined—for example, by exploring different aggregation rules, client selection policies or model architectures—without requiring access to large-scale GPU clusters or custom smart-glasses prototypes. The present report therefore positions FL as an architectural and methodological pillar of the smart-glasses design, while clearly acknowledging that end-to-end federated training of the complete ViT-GPT2 captioning pipeline lies beyond the scope and resources of the current implementation.

3.2 Technology Stack and Cost-Oriented Prototype Design

To complement the conceptual AI pipeline and Federated Learning framework, it is useful to outline a realistic yet resource-aware technology stack for a prototype implementation of AI-enabled smart glasses. The goal of this section is not to prescribe a specific bill of materials (BOM), but rather to identify feasible hardware and software options, and to provide indicative cost estimates under typical pricing. Given that the Saudi Riyal (SAR) is effectively pegged to the US dollar at approximately 3.75 SAR per USD,[41] we express costs in both currencies for clarity.

Hardware platform and sensors.

For a low-cost, educational prototype, a single-board computer such as the Raspberry Pi family is an attractive option. A Raspberry Pi 4 Model B with 4 GB of RAM offers a quad-core 64-bit CPU, integrated Wi-Fi and Bluetooth, and camera interfaces, with a list price from about \$35 internationally,[42, 43] but significantly higher in some local markets (e.g., around 560 SAR for a 4 GB unit on Saudi e-commerce platforms).[44] For ultra-compact, embedded use, the Raspberry Pi Zero 2 W provides a much smaller, low-power board with built-in wireless connectivity at an official price of roughly \$15 (about 55–60 SAR), making it popular for IoT and wearable prototypes.[45, 46]

For the camera, the Raspberry Pi Camera Module 3 (12MP sensor with auto-focus) is a natural choice, with official pricing starting around \$25 (approximately 90–95 SAR) depending on lens field of view and NoIR options.[47, 48] Audio output for visually impaired users can be delivered using either open-ear speakers or a bone-conduction transducer embedded into the glasses frame. Commercial bone-conduction development kits, which typically include the driver electronics and transducer, are available at around \$40–\$50 (roughly 150–190 SAR) per kit.[49, 50] Power can be supplied by a small lithium-ion battery pack (e.g., 5 000–10 000 mAh power bank), with commodity prices in the range of \$15–\$25 (about 55–95 SAR) depending on capacity and brand.

Although the exact mechanical design (3D-printed or off-the-shelf glasses frame, mounting brackets, cabling) is project-dependent, a reasonable academic estimate for structural and miscellaneous components (frame, mounts, connectors, wiring, basic enclosure) is on the order of \$20–\$30 (75–110 SAR). When combined, these elements suggest that a single smart-glasses prototype based on a Pi Zero 2 W could be built for a hardware cost on the order of 150–200 USD (roughly 560–750 SAR), assuming retail pricing and including contingencies for shipping and local mark-ups. A Pi 4-based design would be somewhat more expensive but may offer improved headroom for on-device inference with lighter models.

Software stack (device side).

On the device side, the baseline software stack would consist of a Linux-based operating system (e.g., Raspberry Pi OS) with Python as the main development language. Core AI components include the PyTorch deep-learning framework and the Hugging Face Transformers library to host the VisionEncoderDecoder captioning model (such as `vit-gpt2-image-captioning`), along with pre-processing utilities for the Vision Transformer image encoder.[4, 25] For text-to-speech, the prototype implementation can rely on open-source or freely accessible TTS libraries (e.g., gTTS as used in our experiments) or platform-provided TTS services on a paired smartphone.[39, 40, 51]

In an FL-enabled deployment, a lightweight FL client library (for instance, a Python-based framework running on the device or on a tethered smartphone) would be responsible for: (i) orchestrating local fine-tuning of selected model components using the user’s private data; (ii) serialising updates; and (iii) communicating securely with a remote aggregation server. Given the limited compute and memory budgets of embedded boards, such a client would typically only adapt small adapter layers or low-rank factors, rather than the full ViT–GPT2 model.[30, 34] In practice, for resource-constrained student projects, much of this FL client logic can be emulated on a standard laptop while the glasses prototype focuses on the real-time image capture and audio feedback loop.

Server-side infrastructure and FL orchestration.

On the server side, the minimum viable infrastructure for experimentation with Federated Learning consists of: (i) a modest cloud or on-premise virtual machine capable of running the global model and aggregation logic; and (ii) persistent storage for model checkpoints and logs. Commodity cloud virtual machines such as an AWS

EC2 `t3.small` instance (2 vCPUs, 2 GB RAM) are priced at roughly \$0.02 per hour, corresponding to about \$15 per month (approximately 55–60 SAR) for continuous operation,[52] while a Google Cloud `e2-medium` VM (2 vCPUs, 4 GB RAM) is listed at around \$0.03–\$0.04 per hour, or \$24–\$30 per month (roughly 90–115 SAR), depending on region and discounts.[53, 54] Such instances are sufficient for hosting a central FL coordinator and running small to medium-sized models; heavier captioning models may require periodic access to a GPU-backed instance for initial fine-tuning but can be served on CPU thereafter for research purposes.

From a software perspective, standard Python-based FL frameworks or custom implementations can be used to manage client registration, round-based aggregation (e.g., FedAvg), and logging of training metrics.[5, 30] Secure communication (e.g., TLS) and, where applicable, secure aggregation or differential-privacy mechanisms can be layered on top of the basic orchestration. In the context of the present project, the same server infrastructure that supports the FL simulation (with partitioned datasets and virtual clients on a single machine) can, in principle, be extended in future work to interact with real smart-glasses clients once hardware prototypes and network connectivity are available.

Cost summary and alignment with project scope.

Table-style cost breakdowns inevitably depend on local availability, import duties and component substitutions; nevertheless, the estimates above indicate that a single proof-of-concept smart-glasses unit could plausibly be assembled for well under 1 000 SAR in parts, while server-side FL experimentation can be conducted on low-cost cloud instances or university-provided hardware. This aligns with the educational and exploratory nature of the current work: the technology stack is deliberately chosen to be achievable within student-level resource constraints, while still being representative of the architectural decisions that would arise in a more mature assistive product.

Importantly, the cost analysis reinforces the methodological choices described earlier. Because full-scale on-device federated training of a large Vision–Language model remains computationally expensive, the project focuses on: (i) implementing and evaluating the captioning-and-speech pipeline in a centralised setting; (ii) designing a realistic FL architecture tailored to smart glasses; and (iii) validating FL behaviour through lightweight simulations on modest hardware. Within this scope, the proposed technology stack and indicative costs demonstrate that privacy-aware, AI-enabled smart glasses are technologically and economically plausible, even if some components—such as end-to-end federated captioning—are reserved for future work under more generous resource and deployment conditions.

4 Methodology and Implementation

4.1 NLP Dataset and Preprocessing

As described in Section 2.2, the image captioning component of the project is built on the Flickr8k dataset, which contains approximately 8,000 natural images, each annotated with five human-written captions.[3, 19, 20] This dataset is widely used in early image-captioning work because it is small enough to be manageable in constrained

environments while still capturing a diverse range of everyday scenes. For the purposes of this project, we work with a subset of the Flickr8k images and captions in order to keep the computational requirements compatible with a single GPU in a cloud-based notebook environment.

Raw data and file structure.

The original Flickr8k distribution provides images as JPEG files and captions in a text file in which each line consists of an image filename, a caption index and a caption string.[3] In our implementation, we first organised the images into a dedicated directory and verified that all referenced filenames were present. We then parsed the captions file line by line, extracting the image identifier and the corresponding caption text. Any entries whose image file could not be found in the images directory were discarded at this stage to avoid inconsistencies in later processing. To simplify the training setup, we associated one caption per image in the working subset, rather than using all five captions for each image, and stored the resulting pairs in a tabular format using a CSV file with two main columns: `image_path`, containing the full path to the image on disk, and `caption`, containing the corresponding English description.

Subset selection and basic cleaning.

Because the available GPU memory and training time are limited, we restricted attention to a subset of the available image-caption pairs rather than the full Flickr8k dataset. The subset was sampled from the original training portion of Flickr8k so that the model would still see a varied collection of scenes and objects, while keeping the total number of samples at a level suitable for a student project. Basic cleaning steps were applied during the CSV construction and loading process: rows with missing file paths or empty captions were removed; leading and trailing whitespace was stripped from caption strings; and duplicate rows, if any, were dropped. No aggressive normalisation (such as stemming or stop-word removal) was performed, since the goal is to preserve natural sentence structure and fluency rather than compress the vocabulary.

Integration with the VisionEncoderDecoder pipeline.

To interface the dataset with the ViT-GPT2 VisionEncoderDecoder model, we implemented a custom `ImageCaptionDataset` class in PyTorch. Each instance of the dataset stores a reference to the caption CSV, the Hugging Face image processor for the chosen Vision Transformer encoder, and the tokenizer for the GPT-2 decoder.[4, 25] When a sample with index i is requested, the dataset performs the following steps:

1. Loads the image from `image_path_i` using the Python Imaging Library (PIL) and converts it to RGB.
2. Applies the ViT image processor to resize and normalise the image, yielding a tensor of pixel values with the shape expected by the encoder.
3. Tokenises the caption string `caption_i` using the GPT-2 tokenizer, producing input IDs and an attention mask up to a fixed maximum caption length.
4. Constructs the label sequence for training by copying the input IDs and replacing padding-token positions with `-100`, so that they are ignored by the loss function during optimisation.

This design allows the dataset to return a dictionary containing both the processed image tensor (`pixel_values`) and the caption token sequence (`labels`) for each item, which integrates naturally with PyTorch data loaders and the training loop described in the next subsection.

Train-evaluation usage.

In the experiments conducted for this project, the prepared subset of Flickr8k was primarily used as a training set for fine-tuning the pre-trained `vit-gpt2-image-captioning` model. Additional images that were not included in the sampled training subset were reserved for qualitative inspection and for simple quantitative evaluation measures (e.g., approximate caption-matching accuracy and BLEU-1 scores), as discussed in Chapter 5. This separation ensures that the reported results reflect the model’s ability to generate captions for previously unseen scenes, rather than memorising the training captions.

4.2 Captioning Model Fine-Tuning Setup

The core captioning model used in this project is the `nlpconnect/vit-gpt2-image-captioning` VisionEncoderDecoder model, which combines a Vision Transformer (ViT) image encoder with a GPT-2 language decoder.[4, 25] The model is pre-trained on large-scale image-caption datasets and provides a strong baseline for zero-shot caption generation. In our work, we fine-tune this pre-trained model on a subset of the Flickr8k dataset prepared as described in Section 4.1, in order to adapt it to our specific training distribution and to gain hands-on experience with the practical aspects of captioning-model training under limited resources.

Model initialisation and configuration.

The pre-trained VisionEncoderDecoder model, together with its associated image processor and tokenizer, is loaded from the Hugging Face model hub within a Python environment. The model is moved to a GPU device when available; otherwise, training falls back to CPU with smaller batch sizes. In the fine-tuning configuration used for this project, we retain the full ViT-GPT2 architecture and optimise all parameters jointly, rather than freezing the encoder or decoder. The maximum caption length is set to a fixed number of tokens (e.g., 16) to encourage concise outputs and to keep memory usage under control during training and generation. The tokenizer is configured to pad sequences to this maximum length, and padding positions in the label sequence are masked with `-100` so that they do not contribute to the loss.

Training loop and optimisation.

Fine-tuning is performed using a custom training loop in PyTorch rather than a high-level trainer API. The prepared `ImageCaptionDataset` is wrapped in a `DataLoader` with mini-batch training and shuffling enabled. For each batch, the pixel-value tensor and caption labels are transferred to the selected device, and the model is invoked in

training mode with both arguments:

$$\mathcal{L} = \text{CrossEntropy}(\text{decoder outputs}, \text{labels}), \quad (3)$$

where the loss is computed only over non-masked positions. The loss value is back-propagated through the entire VisionEncoderDecoder, and the model parameters are updated using the AdamW optimiser, which is widely used in transformer-based models.[25]

Given the constraints of a single cloud-hosted GPU, we adopt modest hyperparameters: mini-batch sizes in the range of 4–8 samples (depending on the available memory), a learning rate on the order of 5×10^{-5} , and a small number of epochs (typically one full pass over the sampled training subset). Within each epoch, the optimiser performs a standard sequence of operations: forward pass, loss computation, backward pass, parameter update and gradient reset. Periodic printing of the batch loss is used to monitor optimisation progress and to verify that the model is learning rather than diverging.

Resource-aware training strategy.

Because the goal of this project is not to achieve state-of-the-art captioning scores but rather to obtain a functional, explainable model for integration into a smart-glasses pipeline, we deliberately keep the training schedule short and the hyperparameters conservative. Early stopping and extensive hyperparameter tuning are not employed; instead, we focus on obtaining a stable fine-tuning run that demonstrates clear loss reduction and yields qualitatively reasonable captions on held-out images. This is consistent with the educational context of the work and with the broader system-level perspective, in which the captioning model is one component of a larger assistive architecture rather than an isolated benchmark competitor.

Checkpointing and reuse.

After fine-tuning, the model weights are saved as a reusable checkpoint so that the captioning model can be reloaded in inference-only mode without repeating the training process. In subsequent experiments and demonstrations, this checkpointed model is used as the backbone for the image-to-speech pipeline described in Section 3.1, and for evaluation procedures discussed in Chapter 5. This separation between a one-time fine-tuning phase and repeated inference phases mirrors the expected behaviour in a real-world deployment, where model training would be performed infrequently (e.g., in the cloud or during scheduled updates), while on-device caption generation and speech synthesis would be invoked many times during everyday use.

4.3 Caption Generation and Text-to-Speech Pipeline

Once the VisionEncoderDecoder model has been fine-tuned and saved as a reusable checkpoint, it is integrated into an application-level pipeline that maps images to spoken descriptions. This subsection describes the inference procedure, the simple text post-processing applied in the prototype, and the text-to-speech (TTS) integration used to deliver audio feedback to the user.

Inference procedure for caption generation.

In inference mode, the fine-tuned `vit-gpt2-image-captioning` model is loaded from its checkpoint together with the corresponding image processor and tokenizer.[4] Given an input image, the system first converts it to an RGB format using the Python Imaging Library (PIL) and applies the same pre-processing steps as during training (resize and normalisation via the ViT image processor) to obtain a tensor of pixel values. The model is then invoked in generation mode using the Hugging Face `generate` API, with a fixed maximum length and a small beam width (e.g., beam search with four beams and a maximum of 16 tokens) to produce a concise caption. The output token IDs are finally decoded back into a string using the GPT-2 tokenizer, with special tokens removed.

Text cleaning and post-processing.

The raw caption string produced by the decoder can occasionally contain minor artefacts, such as repeated characters or punctuation (e.g., elongated vowels like “aaaaa” or duplicated punctuation marks) that are not desirable in spoken feedback. To mitigate this, the prototype applies simple rule-based post-processing to the generated caption before passing it to the TTS module. Typical rules include collapsing long sequences of identical characters into a single instance, trimming extraneous whitespace, and enforcing a maximum sentence length. These heuristics are intentionally lightweight, but they suffice to improve the readability and naturalness of the output without introducing complex additional models. More advanced post-processing, such as re-ranking alternative beams based on linguistic quality or conditioning on user preferences, is left as future work.

Text-to-speech integration.

For speech synthesis, the prototype uses a Python-based TTS library that converts the cleaned caption string into an audio file (e.g., an MP3 waveform) that can be played back to the user.[51] In our experiments, we adopt an online service wrapper such as gTTS for convenience, but the conceptual design is compatible with offline neural TTS engines (e.g., Tacotron-2 or FastSpeech-style systems) that could be deployed on a paired smartphone or edge device.[39, 40] The TTS component is configured to use an English voice, with default speaking rate and prosody. In a real deployment targeting visually impaired users, it would be important to support multiple languages, adjustable speaking rates, and robust intelligibility in noisy environments.[8, 17]

Interactive prototype interface.

To demonstrate the end-to-end behaviour of the image-to-speech pipeline, we implemented a simple interactive interface using a web-based UI framework (e.g., Gradio) running inside the development environment. The interface exposes an image upload widget as input and two outputs: a text box displaying the generated caption and an audio player widget for playing back the synthesised speech. Internally, the interface calls a single function that encapsulates the steps above: image pre-processing, caption generation, text post-processing and TTS synthesis. This interactive prototype

does not yet run on dedicated smart-glasses hardware, but it provides a concrete, user-facing demonstration of the core functionality envisioned for the final assistive system and serves as a practical validation of the NLP pipeline described in Section 3.1.

4.4 Federated Learning Prototype Methodology

As discussed in Section 3.2, fully deploying Federated Learning (FL) for a large Vision–Language model on resource-constrained smart-glasses hardware is beyond the scope of the present project. Instead, our practical FL contribution takes the form of a small-scale prototype that simulates the core behaviour of the FedAvg algorithm on a simple neural network and partitioned data in a single notebook environment. The purpose of this prototype is educational: to provide hands-on experience with the mechanics of FL rounds, local training and server-side aggregation under realistic computational constraints, rather than to achieve state-of-the-art performance on a particular benchmark.[5, 30, 32]

Synthetic dataset and client partitioning.

In the prototype, we construct a small supervised classification task with low-dimensional inputs and discrete labels (for example, a synthetically generated dataset or a compact benchmark split), so that training can be performed quickly on a CPU or a single GPU. The dataset is then partitioned into several disjoint subsets, each representing a virtual FL client. For simplicity, the partitions are created by randomly assigning samples to clients, which yields moderately heterogeneous but not adversarially imbalanced data distributions. This setup is sufficient to illustrate the effect of distributing training across multiple clients while keeping the overall implementation straightforward and transparent.

Local training on virtual clients.

We define a lightweight feed-forward neural network as the local model, with a small number of fully connected layers and a standard nonlinearity, implemented in PyTorch. Each virtual client stores its own subset of the training data and a copy of the current global model parameters. During a federated round, a client performs a fixed number of local epochs of stochastic gradient descent (or an Adam-type optimiser) on its local subset, starting from the global weights broadcast by the server at the beginning of the round. At the end of local training, the client returns its updated model parameters and optionally simple statistics such as its local training loss.

Server-side FedAvg loop.

On the server side, we implement a minimal FedAvg loop.[5, 30] The server initialises a global model $w^{(0)}$ and then iterates over several communication rounds. In each round t , it selects all virtual clients (or a subset, in more general scenarios), sends them the current global weights $w^{(t)}$, and waits for their locally updated parameters $w_k^{(t+1)}$. The server then computes a weighted average of these client models to form the next global model $w^{(t+1)}$, where the weights are proportional to the number of samples held by each client. This procedure is repeated for a modest number of rounds, chosen to keep the total runtime manageable in the notebook environment.

Monitoring and interpretation of results.

To monitor the behaviour of the FL prototype, we track simple metrics such as the global training or validation accuracy and loss after each communication round. These metrics are computed by evaluating the aggregated model on a held-out subset of the data that is not used for local training. In the experiments conducted for this project, the accuracy typically improves over the first few rounds and then stabilises, providing qualitative evidence that the FedAvg procedure is working as intended on the chosen toy task. The exact numerical values are less important than the observed trends and the opportunity to connect them to the theoretical description of FL in Section 2.3.1.

Relation to the smart-glasses scenario.

Although the FL prototype operates on a simplified dataset and a small neural network, it mirrors, at a conceptual level, the lifecycle that a federated smart-glasses deployment would follow: a global model initialised centrally, local adaptation on device-specific data, periodic aggregation of updates on a server, and performance evaluation over time.[30, 34] By implementing this prototype under student-level resource constraints, we are able to experiment with key FL hyperparameters (such as the number of local epochs and the number of rounds) and to develop an intuitive understanding of how federated training differs from standard centralised training. The same principles would apply if, in future work, a compact version of the captioning model were trained in a federated manner across a population of smart-glasses users, even though such an extension would require additional computational resources and engineering effort beyond the current project.

5 Results and Discussion

5.1 Captioning Model Behaviour and Qualitative Results

The fine-tuning experiments described in Chapter 4 yield a captioning model that is sufficient for demonstrating the core image-to-speech functionality required by the proposed smart-glasses system. In this subsection, we summarise the main observations from training and from qualitative inspection of generated captions on unseen images from the Flickr8k dataset.

Training behaviour.

During fine-tuning, the batch-wise training loss was monitored over a single epoch on the sampled subset of Flickr8k, using mini-batches in the range of 4–8 images and the AdamW optimiser with a learning rate on the order of 5×10^{-5} . [25] The training loop successfully completed on the available hardware without numerical instability or divergence, indicating that the chosen hyperparameters and batch sizes were compatible with the resource constraints of the cloud-based environment. Although extensive hyperparameter sweeps and multi-epoch training were not attempted, the observed loss values provided a basic sanity check that the model was updating its parameters in response to the image–caption pairs, rather than simply memorising padding tokens or failing to learn.

Qualitative captioning examples.

To assess the practical usefulness of the fine-tuned model, we generated captions for images that were not included in the training subset. These images were drawn from the remaining portion of the Flickr8k dataset and from additional test images, in order to mimic the smart-glasses scenario in which the system encounters scenes it has not seen during training.[3, 19] Qualitative inspection of the outputs showed that the model generally captures the high-level structure of the scene (for example, recognising whether the image depicts people, animals or outdoor environments) and often identifies salient objects and actions in a way that would be useful to a visually impaired user. At the same time, the captions are not perfect: they occasionally omit secondary details, confuse specific object categories, or produce generic descriptions when the visual content is ambiguous or atypical, which is consistent with prior analyses of captioning models on limited datasets.[20]

Text quality and artefacts.

As noted in Section 4.3, the raw caption strings sometimes contained minor artefacts such as repeated characters or slightly awkward phrasing, especially for challenging images. The simple rule-based post-processing applied in the prototype (e.g., collapsing long sequences of identical characters and trimming extraneous whitespace) was effective at removing the most obvious issues without altering the overall meaning of the generated sentence. From an assistive-technology perspective, these imperfections are acceptable for a preliminary prototype, since the primary goal is to convey the main elements of the scene rather than to produce human-level literary descriptions.[8] In future iterations, additional linguistic refinement—for example, re-ranking multiple candidate captions or applying lightweight grammatical correction—could further improve the fluency of the spoken output.

Simple quantitative indicators.

In addition to qualitative inspection, we implemented simple quantitative evaluation procedures based on a held-out set of image-caption pairs. These procedures include computing an approximate exact-match caption accuracy (i.e., the proportion of generated captions that exactly match one of the reference captions) and unigram-based BLEU-1 scores as a coarse measure of word overlap.[19, 20] Given the limited size of the training subset, the short fine-tuning schedule and the inherent variability of natural-language descriptions, these metrics are not expected to reach high values. Rather than optimising aggressively for numerical scores, we use them primarily as sanity checks that the fine-tuned model performs better than random caption generation and that its behaviour is consistent with the qualitative observations described above. Detailed benchmarking on larger datasets and with more sophisticated metrics is left to future work.

5.2 Federated Learning Prototype Results

The small-scale Federated Learning (FL) prototype introduced in Section 4.4 was used to explore how a simple global model evolves when trained with the FedAvg

algorithm on partitioned client data. Although the task and model are deliberately minimal, the prototype provides concrete evidence that federated optimisation can make progress under student-level resource constraints and helps to build intuition for how FL behaves in practice.[5, 30, 32]

Setup recap.

The prototype considers a binary classification problem with low-dimensional synthetic inputs, where each sample is assigned to class 0 or class 1 based on a simple rule applied to its feature vector. A total of 600 samples are generated and evenly split among three virtual clients, each holding 200 samples. The local model on each client is a lightweight feed-forward neural network with one hidden layer and a ReLU nonlinearity, implemented in PyTorch. Training proceeds for a small number of communication rounds (five in our experiments), with each client performing a fixed number of local epochs per round using stochastic gradient descent and cross-entropy loss before sending its updated weights back to the server.

Global accuracy across rounds.

After each communication round, the aggregated global model is evaluated on the full synthetic dataset to obtain a simple measure of global accuracy. Starting from an untrained initialisation, the global model achieves accuracy close to random guessing (around 50%) on the first evaluation, as expected for a balanced binary task. Over the course of five federated rounds, the global accuracy increases steadily to a modest but clearly improved level (approximately 58% in the observed run), and the mean local accuracies on the clients follow a similar upward trend. This behaviour is consistent with the theoretical intuition behind FedAvg: repeated cycles of local optimisation and global averaging allow the model to extract useful information from distributed data, even when each client only performs a small amount of training per round.[5, 30]

Interpretation and limitations.

The absolute accuracy values obtained in this toy setup are not high, and they are not intended to be. The model, dataset and training schedule are deliberately minimal in order to keep computation and runtime low, and no effort was made to tune architecture, learning rate or number of rounds for maximum performance. Within this context, the key observation is not that the model reaches a particular accuracy threshold, but that federated training improves performance above random prediction and exhibits the expected qualitative behaviour (monotonic or near-monotonic improvement over rounds) on distributed data. Comparable small models trained centrally on simple synthetic tasks often achieve similar accuracy levels, so the observed results are reasonable for the chosen configuration.

Relevance to the smart-glasses FL framework.

Despite its simplicity, the FL prototype mirrors several important aspects of the smart-glasses scenario outlined in Section 3.2. A single global model is initialised centrally, cloned to multiple clients, updated locally on client-specific data and aggregated periodically on a server. The experiment demonstrates, in a controlled and inexpensive

setting, that such a training loop can be implemented and that it behaves in line with the FedAvg literature. In future work, the same code structure could be extended to more realistic datasets and to compact vision or vision–language models, subject to the availability of additional computational resources. For the present project, however, the prototype serves its intended purpose as an educational and methodological bridge between the formal description of FL and the conceptual design of a federated smart-glasses system.[30, 34]

5.3 Discussion

The results obtained in the captioning and Federated Learning components can now be interpreted in light of the project objectives stated in Chapter 1. Broadly, the work aimed (i) to propose a conceptual framework for integrating Federated Learning into AI-enabled smart glasses, (ii) to implement and study an image captioning pipeline suitable for visual assistance, and (iii) to develop a resource-aware understanding of how FL behaves in practice under realistic constraints.

Federated Learning objectives.

From the FL perspective, the project set out to develop a clear conceptual framework for how smart glasses could participate as clients in a federated ecosystem and to deepen understanding of core FL mechanisms, challenges and trade-offs.[30, 34] These goals are addressed primarily through the conceptual design in Chapter 3 and the small-scale prototype in Chapters 4 and 5. The architecture diagrams and technology-stack analysis show how a population of AI-enabled smart glasses could, in principle, perform local updates on-device while a central server aggregates model parameters, thereby preserving user privacy and reducing raw data transmission. The toy FL prototype does not attempt to train a full Vision–Language model but successfully demonstrates the FedAvg lifecycle on a simplified task, illustrating how local epochs, client partitions and communication rounds interact. In this sense, the project achieves its FL objectives at an educational and methodological level, while explicitly deferring full-scale federated captioning to future work.

NLP and captioning objectives.

On the NLP side, the project aimed to implement a practical image captioning pipeline that connects visual perception to natural-language descriptions and speech output for visually impaired users. This objective is met by fine-tuning the `vit-gpt2-image-captioning` model on a curated subset of Flickr8k, integrating it with a text-to-speech module and exposing the combined functionality through an interactive interface.[3, 4, 8] The qualitative results show that the model can generate concise, reasonably accurate descriptions of unseen images, and the TTS integration confirms that these descriptions can be delivered as audio feedback with minimal latency in a prototype setting. While the evaluation metrics remain modest, they are consistent with the limited training budget and the small dataset, and they suffice to demonstrate that the captioning component is viable as the core NLP engine of the envisioned smart-glasses system.

System-level and interdisciplinary objectives.

A further objective of the project was to adopt a system-level view that connects literature, algorithms, architecture and approximate cost into a coherent narrative suitable for a senior capstone context. The deep literature review synthesises recent work on assistive smart glasses, image captioning and Federated Learning, highlighting both opportunities and gaps.[19, 20, 32] The conceptual framework and technology-stack analysis translate these insights into an implementable design, including concrete hardware and deployment options that fall within realistic student-level budgets. At the same time, the implemented prototypes—the captioning-and-speech pipeline and the FL sandbox—provide hands-on evidence that the proposed ideas are not purely theoretical. Overall, although the project intentionally operates at a modest computational scale, it successfully aligns its practical outcomes with the stated objectives and lays a grounded foundation for a future senior project that could extend the same concepts to more advanced models and real smart-glasses hardware.

6 Conclusion and Future Work

6.1 Conclusion

This project investigated how AI-enabled smart glasses could support visually impaired users by combining image captioning with a privacy-aware Federated Learning (FL) perspective. The work brought together three main strands: a literature-grounded conceptual framework for assistive smart-glasses and FL; a practical image-to-speech pipeline based on a Vision–Language captioning model; and a small-scale FL prototype that illustrates the dynamics of federated optimisation under realistic resource constraints.

On the NLP side, a pre-trained VisionEncoderDecoder model (`vit-gpt2-image-captioning`) was fine-tuned on a curated subset of the Flickr8k dataset to generate concise English descriptions of images.[3, 4, 19, 20] The fine-tuned model was integrated into an image-to-speech pipeline by adding simple post-processing and a text-to-speech component, and exposed through an interactive interface. Qualitative inspection confirmed that the system can produce useful high-level descriptions of unseen images and deliver them as audio feedback, demonstrating that it is a viable core NLP engine for an assistive smart-glasses scenario, even when trained with modest computational resources.[8]

From the FL perspective, the project did not attempt to train the Vision–Language model in a federated manner, but instead focused on developing a clear architectural and methodological understanding of how FL could be integrated into smart glasses. A conceptual client–server framework was proposed in which smart glasses act as FL clients that retain their local image data and periodically send model updates to a coordinating server, which aggregates them using FedAvg.[5, 30, 34] This framework was complemented by a lightweight FL prototype on a toy classification task, showing that a simple global model can improve over several federated rounds when data is partitioned across multiple virtual clients. Together, these components demonstrate that FL is both conceptually compatible with assistive smart-glasses deployments and practically implementable at a small experimental scale.

Beyond individual algorithms, the project adopted a system-level view by mapping the literature onto a concrete smart-glasses architecture and outlining a realistic technology stack and cost envelope for a proof-of-concept prototype. The resulting narrative connects recent research on assistive wearables, image captioning and Federated Learning with implementation details and resource-aware design choices, thereby laying a grounded foundation that can be expanded in a future senior project.

6.2 Limitations

Several limitations must be acknowledged when interpreting the outcomes of this work. First, the image captioning component is trained on a reduced subset of Flickr8k with a single caption per image and a short fine-tuning schedule. As a result, the model’s performance is constrained by limited data diversity, training time and hyperparameter exploration, and it is not intended to compete with state-of-the-art captioning systems trained on larger datasets such as MS COCO.[19, 20] The evaluation metrics used—including simple exact-match accuracy and BLEU-1 scores on held-out data—are likewise modest and serve primarily as sanity checks rather than exhaustive benchmarks.

Second, the FL experiments are deliberately restricted to a toy synthetic classification task and a very small neural network. Although this setup is suitable for understanding the mechanics of FedAvg and for running experiments on student-level hardware, it does not capture the full complexity of real-world FL deployments with non-IID data, intermittent connectivity and heterogeneous devices.[30, 32] No real smart-glasses hardware was built in this phase of the work, and the conceptual FL framework has not yet been validated with actual on-device training or communication constraints.

Third, the current prototype focuses on English-only captions and TTS, without addressing multilingual support, user-specific personalisation or extensive usability testing with visually impaired users. These aspects are crucial for a deployable assistive system but lie beyond the scope of the present project. Finally, all experiments are conducted in controlled notebook environments, and no end-to-end latency, energy consumption or robustness measurements are reported for embedded or mobile hardware.

6.3 Future Work and Link to Senior Project

The present project can be viewed as a first phase in a broader smart-glasses research agenda, providing conceptual and experimental groundwork that a future senior project can extend. One natural line of future work is to strengthen the captioning component by training on larger and more diverse datasets, exploring more compact or efficient Vision–Language architectures, and performing more systematic evaluation with richer metrics. In parallel, the text-to-speech subsystem could be upgraded to support multiple languages, adjustable speaking rates and offline neural TTS engines running on a paired smartphone or edge device, thereby improving the usability of the system for visually impaired users in real-world environments.[8]

On the FL side, a key direction is to move from toy synthetic tasks towards more realistic, privacy-relevant datasets and models, while maintaining compatibility with available hardware. This could include experimenting with small vision or captioning models in an FL setting, simulating non-IID client data distributions, and investigating the impact of varying the number of clients, local epochs and aggregation strategies.[30, 34?] As computational resources permit, elements of the conceptual FL framework for smart glasses—such as client-side update logic, secure communication and hierarchical aggregation via edge servers—could be prototyped and evaluated more thoroughly.

Finally, at the system level, a future senior project could focus on building and evaluating a physical smart-glasses prototype based on the technology stack outlined in Section 3.3, using a lightweight captioning model as the on-device or phone-assisted inference engine and leveraging the FL insights gained here to guide the design of privacy-aware update mechanisms. In this way, the current work serves as a conceptual and methodological foundation, while the senior project would represent a second phase that emphasises hardware integration, user-centred evaluation and incremental movement towards a deployable assistive device.

References

- [1] Envision Glasses for Blind and Low-Vision Users. <https://www.letsenvision.com/glasses/home>. Accessed: 2025-12-01 (2025)
- [2] Verge, T.: These Smart Glasses Use AI to Help Low-Vision Users. Accessed: 2025-12-01
- [3] Hodosh, M., Young, P., Hockenmaier, J.: Framing image description as a ranking task: Data, models and evaluation metrics. *Journal of Artificial Intelligence Research* **47**, 853–899 (2013)
- [4] Hsieh, Y.-S., contributors: nlpconnect/vit-gpt2-image-captioning. <https://huggingface.co/nlpconnect/vit-gpt2-image-captioning>. Accessed: 2025-12-01
- [5] McMahan, H.B., Moore, E., Ramage, D., Hampson, S., Arcas, B.: Communication-efficient learning of deep networks from decentralized data. In: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)* (2017)
- [6] Truong, N., Sun, K., Wang, S., Guitton, F., Massacci, F.: Privacy preservation in federated learning: An insightful survey. *Computer Security* (2021). Available via ScienceDirect
- [7] Zhan, S., *et al.*: Federated learning architectures for privacy-preserving collaborative ai: A review. *Electronics* **14**(13), 2512 (2025)
- [8] Naayini, H., *et al.*: Ai-powered assistive technologies for visual impairment. *arXiv abs/2503.15494* (2025)

- [9] Li, H., et al.: Federated learning for data security and privacy-preserving analytics in smart IoT systems. *Artificial Intelligence Review* (2024). Online first
- [10] Mekonnen, F.Y., Al Bataineh, M.F., Abu Abdoun, D., Serag, A., Tamiru, K.T., Abula, W., Darota, S.: Development of a fully autonomous offline assistive system for visually impaired individuals: A privacy-first approach. *Sensors* **25**(19), 6006 (2025) <https://doi.org/10.3390/s25196006>
- [11] Samavati, F., et al.: Assistive technologies for the visually impaired: A review. *Cureus Journal of Engineering* (2025). Online ahead of print
- [12] Zhang, X., Huang, X., Ding, Y., Long, L., Li, W., Xu, X.: Advancements in smart wearable mobility aids for visual impairments: A bibliometric narrative review. *Sensors* **24**(24), 7986 (2024) <https://doi.org/10.3390/s24247986>
- [13] Envision Technologies: Envision Glasses. <https://www.letsenvision.com/envision-glasses>. Accessed December 2025 (2025)
- [14] Author, F., Author, S.: Ai enabled smart glasses for visually impaired persons. *International Journal of Innovative Science and Research Technology* **10**(3), 870–878 (2025). Paper ID: IJISRT25MAR667
- [15] Author, F., Author, S.: Ai powered smart glasses for the visually impaired. *International Journal of Creative Research Thoughts* **13**(3), 380–390 (2025). Paper ID: IJCRT2503499
- [16] Author, F., Author, S.: Gen-ai powered smart glasses for visually impaired people. *International Journal of Innovative Research in Technology* **11**(11), 5470–5478 (2025). Paper ID: IJIRT176457
- [17] Udayakumar, D., Gopalakrishnan, S., Raghuram, A., Kartha, A., Krishnan, A.K., Ramamirtham, R., Muthangi, R., Raju, R.: Artificial intelligence-powered smart vision glasses for the visually impaired. *Indian Journal of Ophthalmology* **73**(Suppl 3), 492–497 (2025) <https://doi.org/10.4103/IJO.IJO.1621.24>
- [18] Wang, S., Zhang, W., Gunter, T.D., *et al.*: Integrating ai-powered smart glasses into digital health management for proactive healthcare solutions: A systematic literature review. *npj Digital Medicine* **8**, 410 (2025) <https://doi.org/10.1038/s41746-025-01715-x>
- [19] Hossain, M.Z., Sohel, F., Shiratuddin, M.F., Laga, H.: A comprehensive survey of deep learning for image captioning. *arXiv preprint arXiv:1810.04020* (2018)
- [20] Sharma, H., Padha, D.: A comprehensive survey on image captioning: From hand-crafted to deep learning-based techniques, a taxonomy and open research issues. *Artificial Intelligence Review* **56**, 13619–13661 (2023) <https://doi.org/10.1007/s10462-023-10488-2>

- [21] Vinyals, O., Toshev, A., Bengio, S., Erhan, D.: Show and tell: A neural image caption generator. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 3156–3164 (2015)
- [22] Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R., Bengio, Y.: Show, attend and tell: Neural image caption generation with visual attention. In: Proceedings of the 32nd International Conference on Machine Learning (ICML), pp. 2048–2057 (2015)
- [23] Anderson, P., He, X., Buehler, C., Teney, D., Johnson, M., Gould, S., Zhang, L.: Bottom-up and top-down attention for image captioning and visual question answering. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 6077–6086 (2018)
- [24] Sarto, S., *et al.*: Image captioning evaluation in the age of multimodal LLMs. In: Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI) (2025). Image captioning evaluation metrics for modern models
- [25] Kumar, A.: The illustrated image captioning using transformers. ankur3107.github.io (2022)
- [26] Gurari, D., *et al.*: VizWiz-Captions: A Dataset for Image Captioning for People Who Are Blind. <https://vizwiz.org/tasks-and-datasets/image-captioning/>. Accessed December 2025 (2018)
- [27] Yu, L., Nikandrou, M., Jin, J., Rieser, V.: Quality-agnostic image captioning to safely assist people with vision impairment. In: Proceedings of the 32nd International Joint Conference on Artificial Intelligence (IJCAI), pp. 6284–6292 (2023)
- [28] Nurgaliyev, S.: Image Captioning for the Visually Impaired and Blind: A Recipe for Low-Resource Languages. TechRxiv preprint. Preprint on low-resource image captioning for assistive devices (2023). <https://www.techrxiv.org>
- [29] Mroueh, Y.: Image Captioning as an Assistive Technology. IBM Research Blog. Accessed December 2025 (2018). <https://research.ibm.com/blog/image-captioning-assistive-technology>
- [30] Kairouz, P., McMahan, H.B., *et al.*: Advances and open problems in federated learning. Foundations and Trends in Machine Learning **14**(1–2), 1–210 (2021) <https://doi.org/10.1561/22000000083>
- [31] Truong, N., Sun, K., Wang, S., Guitton, F., Guo, Y.-k.: Privacy preservation in federated learning: An insightful survey from the GDPR perspective. Computers & Security **110**, 102402 (2021) <https://doi.org/10.1016/j.cose.2021.102402>
- [32] Reddy, K.D., *et al.*: A comprehensive survey on federated learning techniques for

- the internet of medical things. *Healthcare* **11**(6), 806 (2023) <https://doi.org/10.3390/healthcare11060806>
- [33] Abbas, S.R., *et al.*: Federated learning in smart healthcare: Architectures, applications and challenges. *Healthcare* **12**(24), 2587 (2024) <https://doi.org/10.3390/healthcare12242587>
 - [34] Zhan, S., Huang, L., Luo, G., Zheng, S., Gao, Z., Chao, H.-C.: A review on federated learning architectures for privacy-preserving AI: Lightweight and secure cloud–edge–end collaboration. *Electronics* **14**(13), 2512 (2025) <https://doi.org/10.3390/electronics14132512>
 - [35] Guan, H., Yap, P.-T., Bozoki, A., Liu, M.: Federated learning for medical image analysis: A survey. *Pattern Recognition* **150**, 110424 (2024) <https://doi.org/10.1016/j.patcog.2024.110424>
 - [36] Nguyen, D.C., Pham, Q.-V., Pathirana, P.N., Ding, M., Seneviratne, A., Lin, Z., Hwang, W.-J.: Federated learning for smart healthcare: A survey. *ACM Computing Surveys* **55**(3), 1–37 (2022) <https://doi.org/10.1145/3501296>
 - [37] Li, L., *et al.*: Cloud–edge–end collaborative privacy-preserving federated learning in non-iid environments. *Sensors* **24**(24), 8028 (2024) <https://doi.org/10.3390/s24248028>
 - [38] Wu, Z., Sun, S., Wang, Y., Liu, M., Xu, K., Pan, O., Gao, B., Wen, T.: Beyond model scale limits: End–edge–cloud federated learning with self-rectified knowledge agglomeration. *arXiv preprint arXiv:2501.00693* (2025)
 - [39] Shen, J., *et al.*: Natural tts synthesis by conditioning Wavenet on Mel-Spectrogram predictions. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 4779–4783 (2018)
 - [40] Ren, Y., *et al.*: FastSpeech: Fast, robust and controllable text to speech. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2019)
 - [41] Investopedia Editors: Saudi Riyal (SAR): Exchange Rate Peg to the US Dollar. <https://www.investopedia.com/terms/forex/s/sar-saudi-riyal.asp>. Accessed December 2025. The Saudi Riyal is pegged at approximately 3.75 SAR per USD (2025)
 - [42] Raspberry Pi Ltd.: Raspberry Pi 4 Model B. <https://www.raspberrypi.com/products/raspberry-pi-4-model-b/>. Accessed December 2025 (2019)
 - [43] The Pi Hut: Raspberry Pi 4 Model B Technical Specifications. <https://thepihut.com/products/raspberry-pi-4-model-b>. Accessed December 2025 (2019)
 - [44] Amazon.sa: Raspberry Pi 4 Model B (4GB) on Amazon.sa. <https://www.amazon>.

- sa/. Example pricing around 560 SAR for 4GB model, accessed December 2025 (2025)
- [45] Raspberry Pi Ltd.: Raspberry Pi Zero 2 W. <https://www.raspberrypi.com/products/raspberry-pi-zero-2-w/>. Official price around \$15, accessed December 2025 (2021)
 - [46] Pimoroni: Raspberry Pi Zero 2 W for Embedded and IoT Projects. <https://shop.pimoroni.com/products/raspberry-pi-zero-2-w>. Accessed December 2025 (2021)
 - [47] Raspberry Pi Ltd.: Raspberry Pi Camera Module 3. <https://www.raspberrypi.com/products/camera-module-3/>. Starting price around \$25, accessed December 2025 (2023)
 - [48] Pimoroni: Raspberry Pi Camera Module 3 Specifications. <https://shop.pimoroni.com/products/raspberry-pi-camera-module-3>. Accessed December 2025 (2023)
 - [49] DFRobot: KIT0126 Bone Conduction Kit. <https://abra-electronics.com/robotics-embedded-electronics/voice-ics-and-modules-en/kit0126-bone-conduction-kit.html>. Approximate price around \$49, accessed December 2025 (2024)
 - [50] Core Electronics: Bone Conduction Kit (DFRobot) – Product Description. <https://core-electronics.com.au/bone-conduction-kit.html>. Accessed December 2025 (2024)
 - [51] Contributors, P.: gTTS: Python Library and CLI Tool to Interface with Google Translate’s Text-to-Speech API. <https://pypi.org/project/gTTS/>. Accessed December 2025 (2025)
 - [52] Vantage: AWS EC2 t3.small Pricing and Specifications. <https://instances.vantage.sh/aws/ec2/t3.small>. On-demand price around \$0.0208 per hour, accessed December 2025 (2025)
 - [53] Google Cloud: Google Compute Engine VM Pricing: e2-medium. <https://cloud.google.com/compute/vm-instance-pricing>. Base on-demand pricing for e2-medium, accessed December 2025 (2025)
 - [54] PriceTimeline: Google Cloud Compute e2-medium Price History. <https://pricetimeline.com/data/web-dev/google-cloud-price-history>. Reports e2-medium around \$24–\$30 per month, accessed December 2025 (2025)